# HARDWARE HACKATHON
# ROUND 1

CareComm Bot: Smart Hospital Communication bot

## TEAM ID:QXHH-20

## TEAM NAME: OMNITRIX

# PROBLEM UNDERSTANDING & PROPOSED IDEA

**PROBLEM STATEMENT:**

- In busy hospitals, communicating medicine and supply requests from patient rooms to the pharmacy often faces delays. Small to medium hospitals lack the expensive, automated infrastructure to handle these logistics efficiently, which increases the manual workload for nurses.

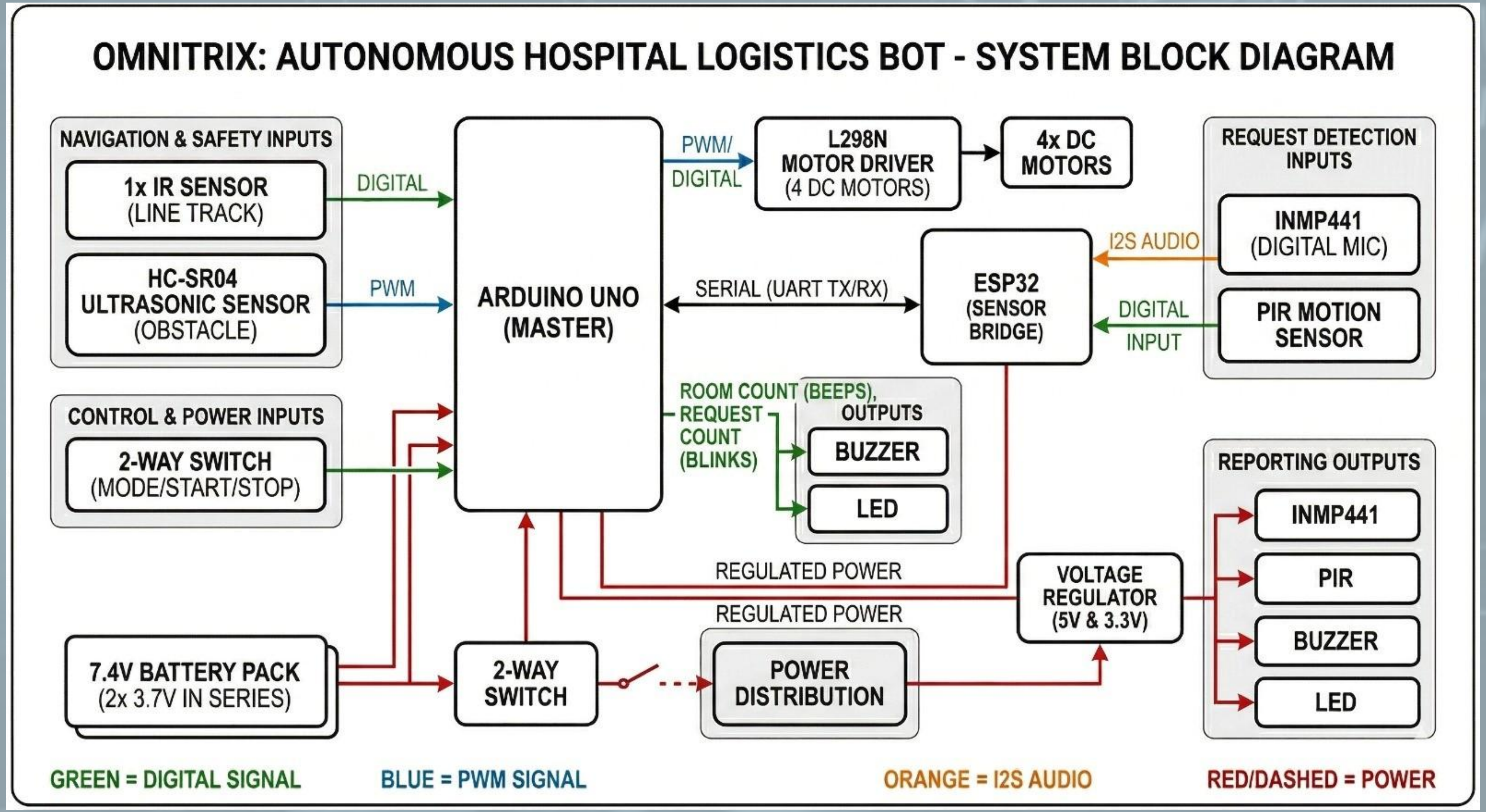
**PROPOSED SOLUTION:**

- A low-cost, line-following autonomous robot that navigates hospital wards, stops at patient rooms to detect physical or audio requests, and returns to the pharmacy to automatically report the total tallies.

**WHAT MAKES IT UNIQUE**:

- It operates completely independently. Instead of relying on expensive hospital Wi-Fi networks, central servers, or complex indoor mapping, our bot uses a reliable hardware-loop (following a physical path). It is a simple, plug-and-play solution that any clinic can deploy immediately.

# SYSTEM ARCHITECTURE



# WORKING

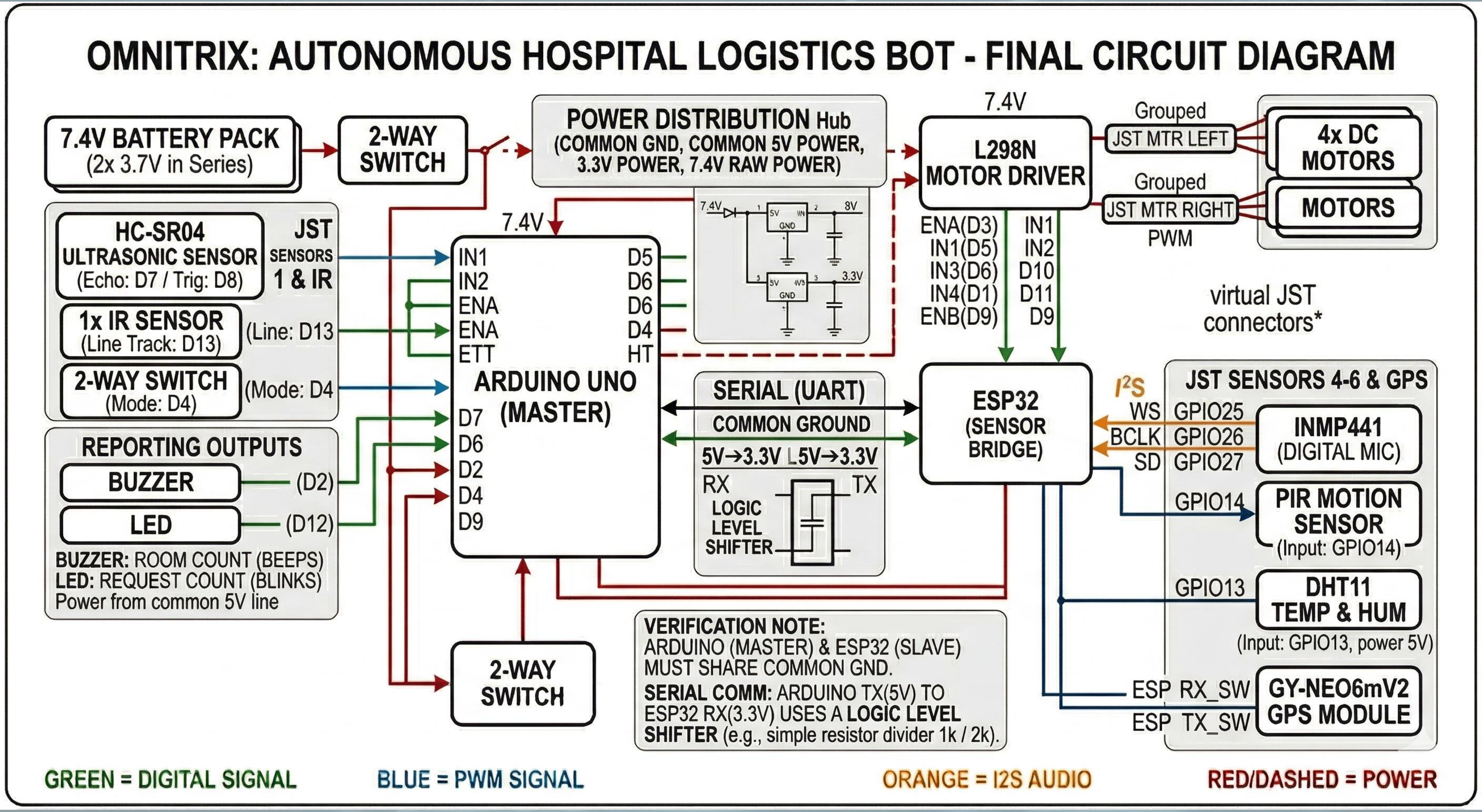
**How robot navigates:** The Arduino reads the IR sensors to follow a continuous line on the floor. It uses the HC-SR04 ultrasonic sensor to halt if a person or obstacle blocks its path.

**How request is detected:** When the bot stops at a room marker, the ESP activates the PIR sensor and INMP441 microphone to detect a patient's hand wave or voice command (like a clap/call).

**How data is processed:** The ESP acts as a helper, sending a simple "Yes/No" signal over Serial to the Arduino if a request is detected. The Arduino updates its internal room and request counters

**How output is given:** Upon reaching the final pharmacy station, the Arduino triggers the LED and Buzzer to output the final data.

# CIRCUIT DIAGRAM



# INNOVATION, FEASIBILITY & OUTPUT

**What innovative features being added?**

- **Dual-Controller Setup:** We use the Arduino for stable, real-time motor control, and the ESP strictly as a dedicated "sensor reader" to handle the complex digital microphone data. This prevents the bot from lagging while driving.
- **Multi-Modal Detection:** We use both motion (PIR) and sound (INMP441) to ensure a request is accurately captured.

**Feasibility:**

- Highly feasible. By relying on proven IR line-following mechanics for the core movement rather than complex AI mapping, we guarantee the robot will reliably complete its physical route. The separation of tasks between Arduino and ESP makes the code simple and easy to debug.

**Expected Output & How will robot behave?**

- The robot will autonomously follow the ward path, pausing completely at each room marker for 3-5 seconds to check for a request, and automatically resume driving until it reaches the end.

**How will requests be counted/displayed?**

At the pharmacy station:

- **LED Blinks** = Total number of requests detected.
- **Buzzer Beeps** = Total number of rooms visited.